
CAP 5516

Medical Image Computing

(Spring 2025)

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Lecture 18

Introduction of Generative Adversarial Networks (GANs)

Generative adversarial nets

I Goodfellow, J Pouget-Abadie, M Mirza, B Xu, D Warde-Farley, S Ozair, A Courville...

Advances in neural information processing systems, 2014 · proceedings.neurips.cc

Abstract

We propose a new framework for estimating generative models via adversarial nets, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G . The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D , a unique solution exists, with G

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Goodfellow, Ian, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. "Generative adversarial nets." Advances in neural information processing systems 27 (2014).



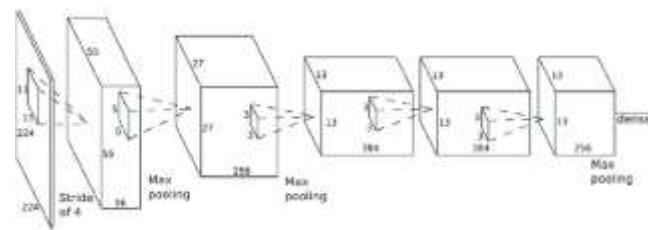
Generative Adversarial Networks (GANs)

Generative

Adversarial

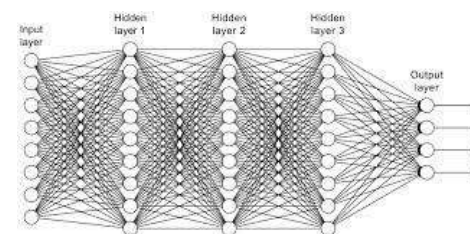
Networks

Generates data
(Create fake data)

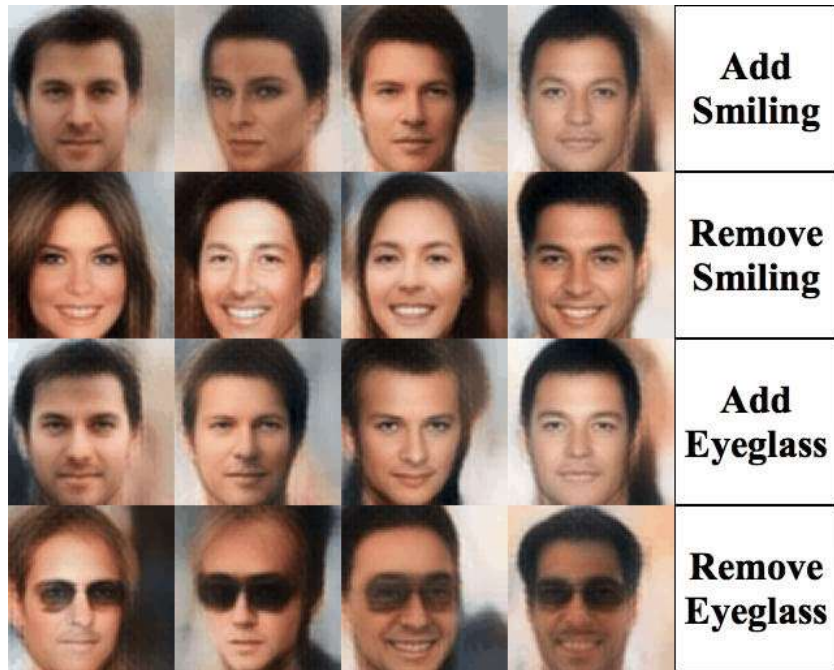


Generator and discriminator,
Each competing to win

Generator trying to fake
Discriminator trying not to be fooled



Applications of GANs

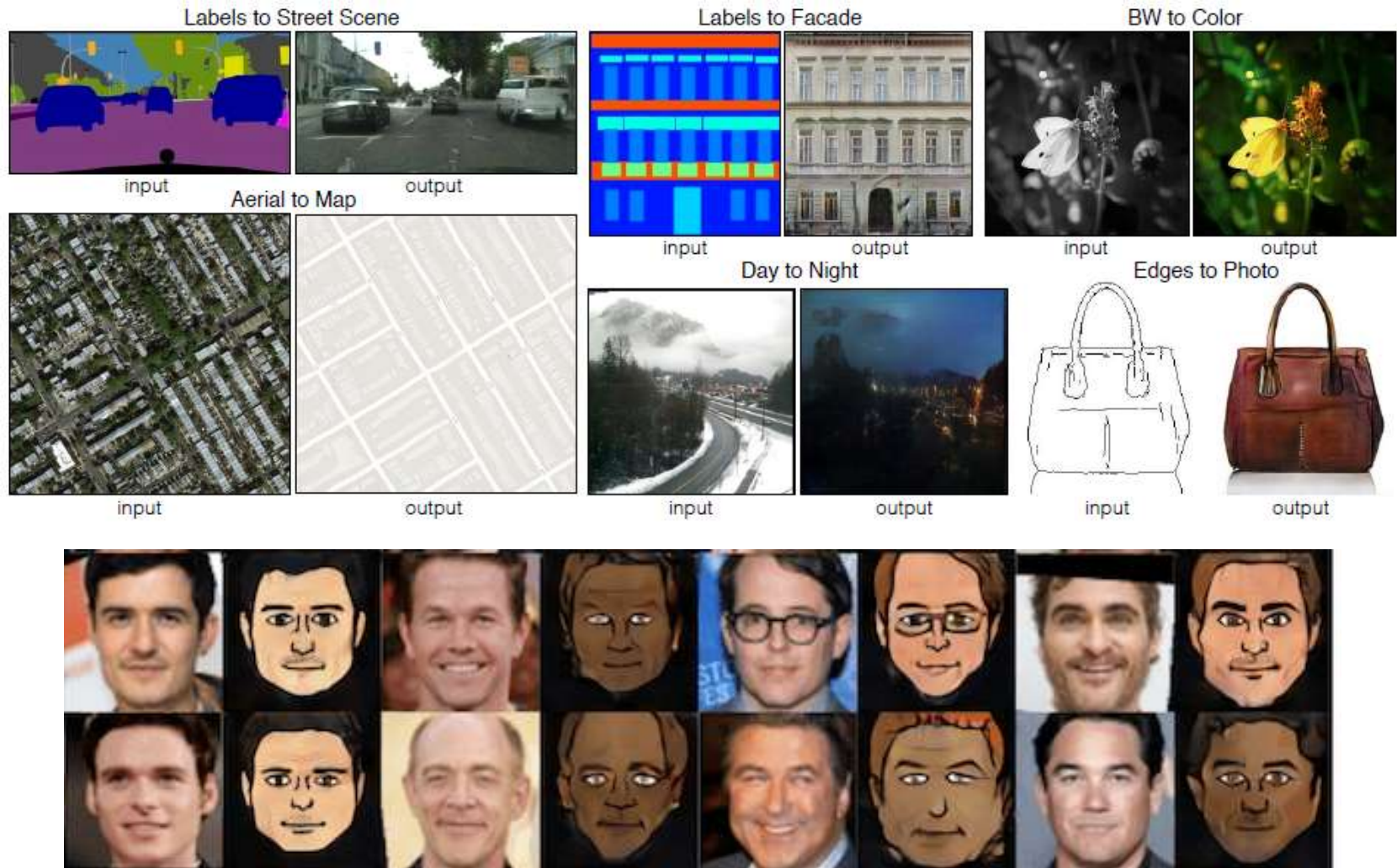


Generate Examples for Image Datasets,
Generate Realistic Photographs



Applications of GANs

Image-to-Image Translation



Isola, Phillip, et al. "Image-to-image translation with conditional adversarial networks." Proceedings of the IEEE conference on computer vision and pattern recognition. 2017.



Applications of GANs

this small bird has a pink breast and crown, and black primaries and secondaries.



this magnificent fellow is almost all black with a red crest, and white cheek patch.



Text-to-Image Translation

the flower has petals that are bright pinkish purple with white stigma



this white and yellow flower have thin white petals and a round yellow stamen



Zhang, Han, et al. "Stackgan: Text to photo-realistic image synthesis with stacked generative adversarial networks." Proceedings of the IEEE international conference on computer vision. 2017.



Applications of GANs

Real image



Reconstructed images



Blonde ↑

Bangs ↑

Smile ↑

Male ↑

Photograph Editing

Perarnau, G., Van De Weijer, J., Raducanu, B., & Álvarez, J. M. (2016). Invertible conditional gans for image editing. arXiv preprint arXiv:1611.06355.

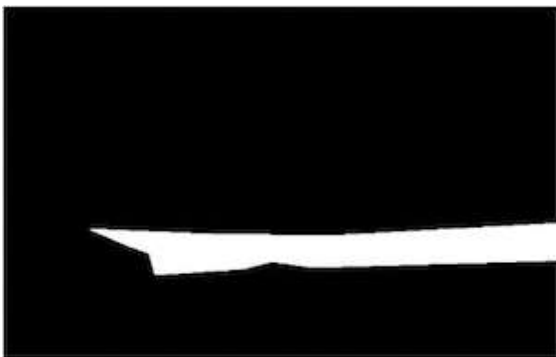


Photo Blending

Wu, Huikai, et al. "Gp-gan: Towards realistic high-resolution image blending." Proceedings of the 27th ACM international conference on multimedia. 2019.

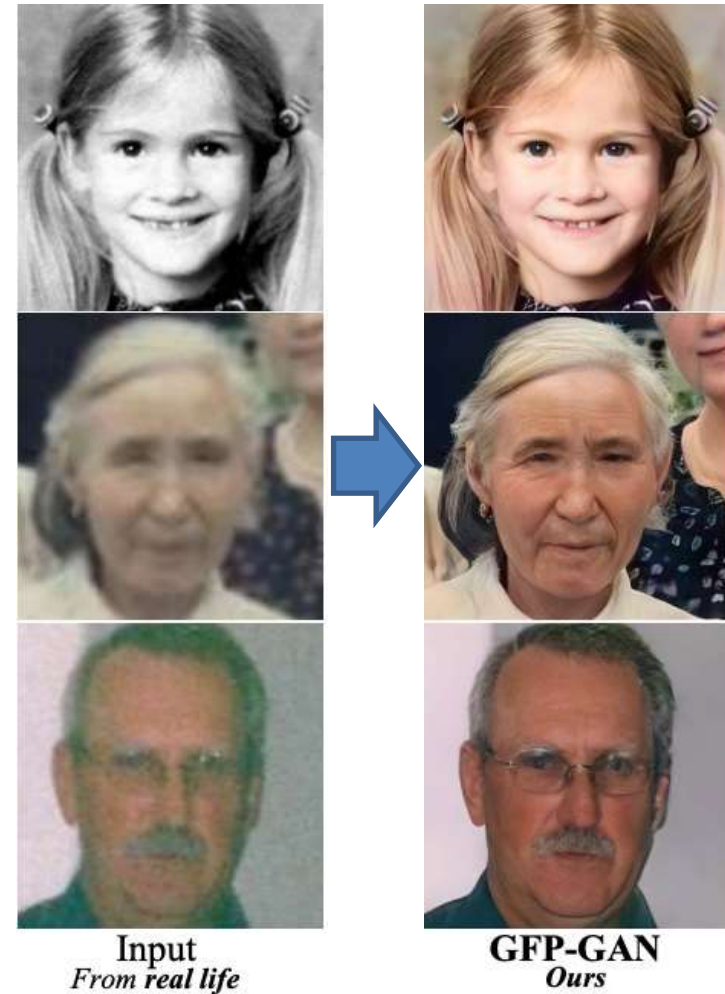


Applications of GANs

Image Recovery, Image Enhancement



Menon, Sachit, et al. "Pulse: Self-supervised photo upsampling via latent space exploration of generative models." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2020.

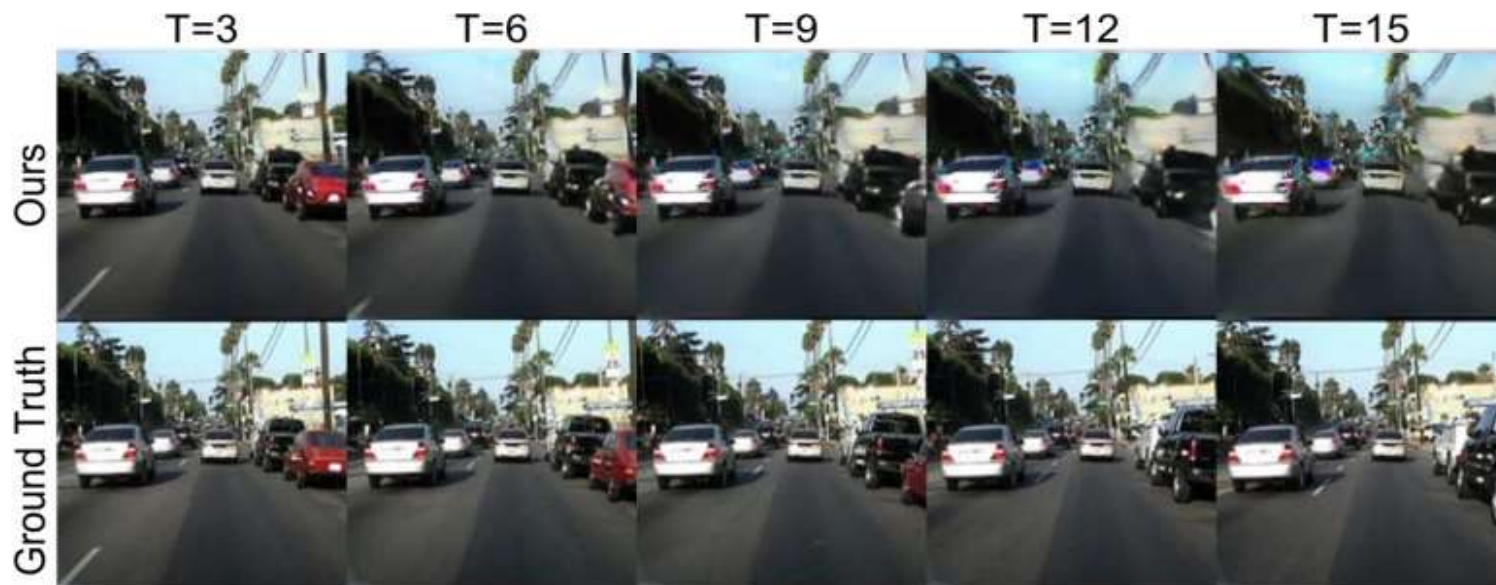


Wang, Xintao, et al. "Towards real-world blind face restoration with generative facial prior." CVPR 2021.

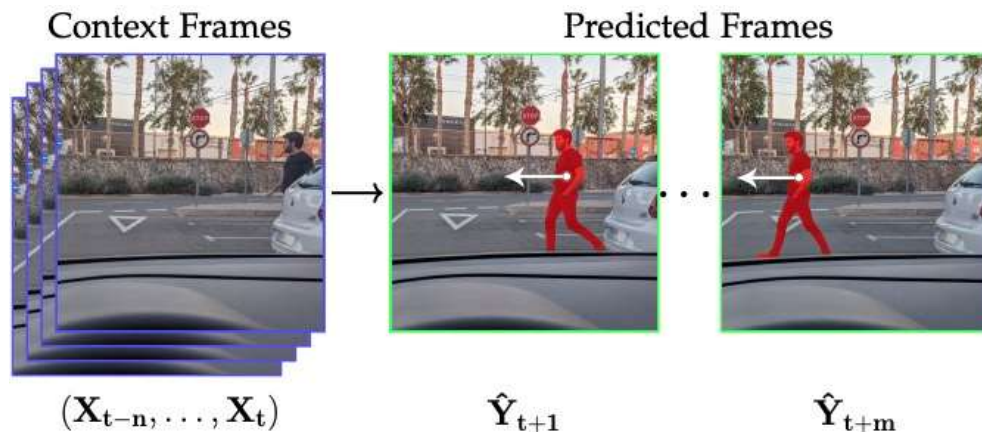


Applications of GANs

Video frames prediction



Kwon, Yong-Hoon, and Min-Gyu Park. "Predicting future frames using retrospective cycle gan." CVPR 2019.

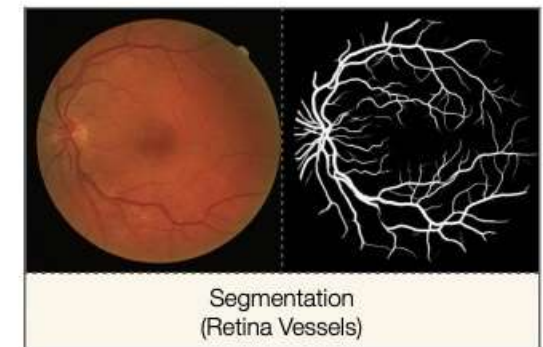
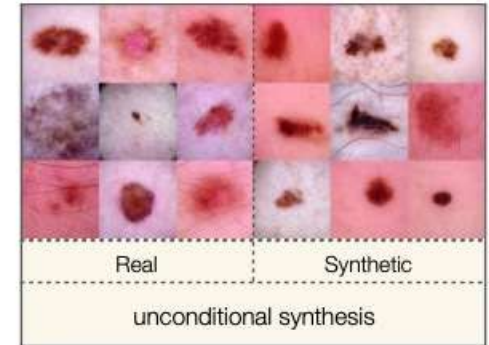
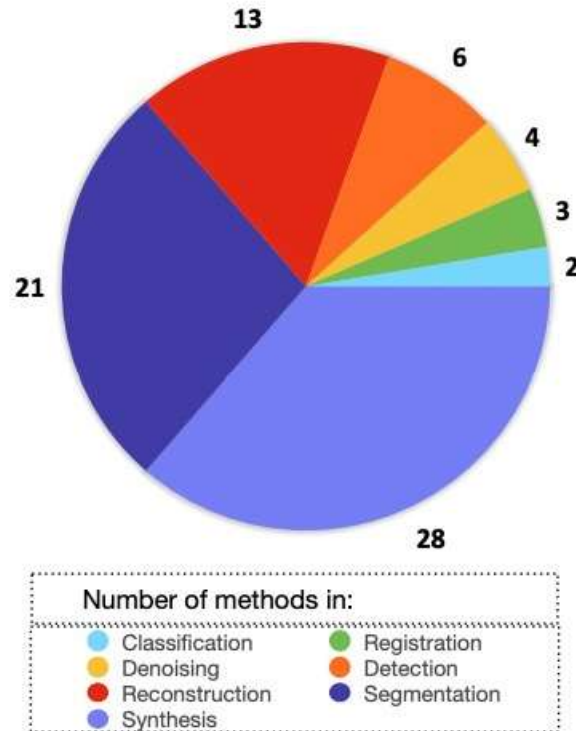
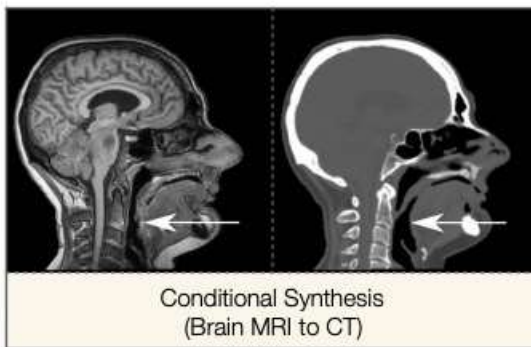
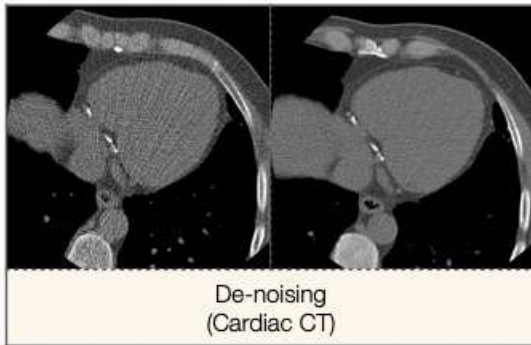
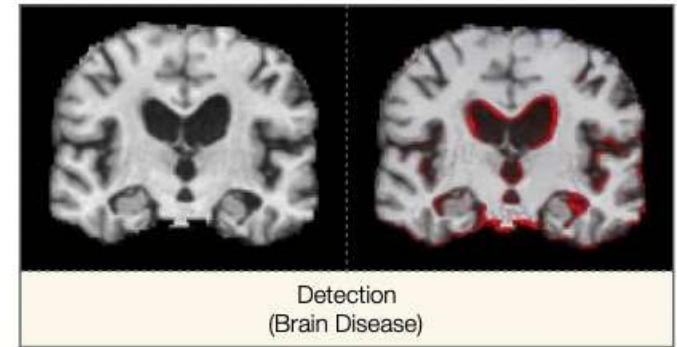
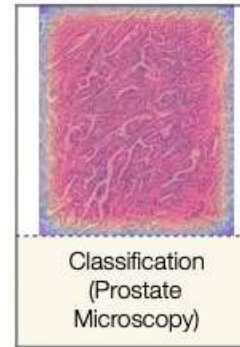
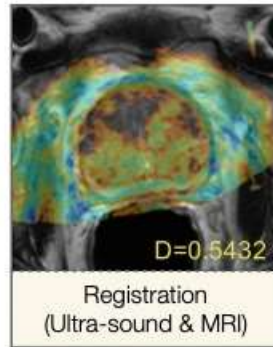
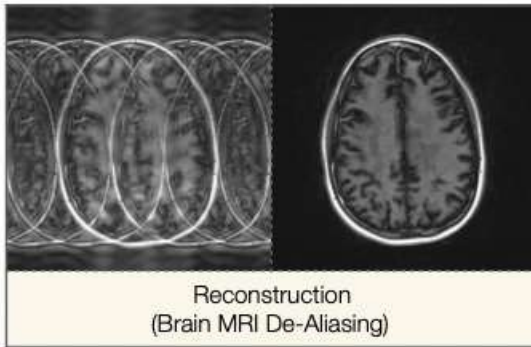


Will the car hit the pedestrian?

Oprea, Sergiu, et al. "A review on deep learning techniques for video prediction." IEEE Transactions on Pattern Analysis and Machine Intelligence (2020).



GANs for Medical Imaging



Kazemina, Salome, et al. "GANs for medical image analysis." Artificial Intelligence in Medicine 109 (2020): 101938.



High level overview of GAN

Output: Sample from training distribution



Generator Network

Input: Random noise

z

Slide Credit: Feifei Li, Justin Johnson, Serena Yeung, Ranjay Krishna, Danfei Xu

High level overview of GAN

- Training GAN: Two-player game

Player 1: Generator

Player 2: Discriminator

Generator network: try to fool the discriminator by generating real-looking images (Art forger)

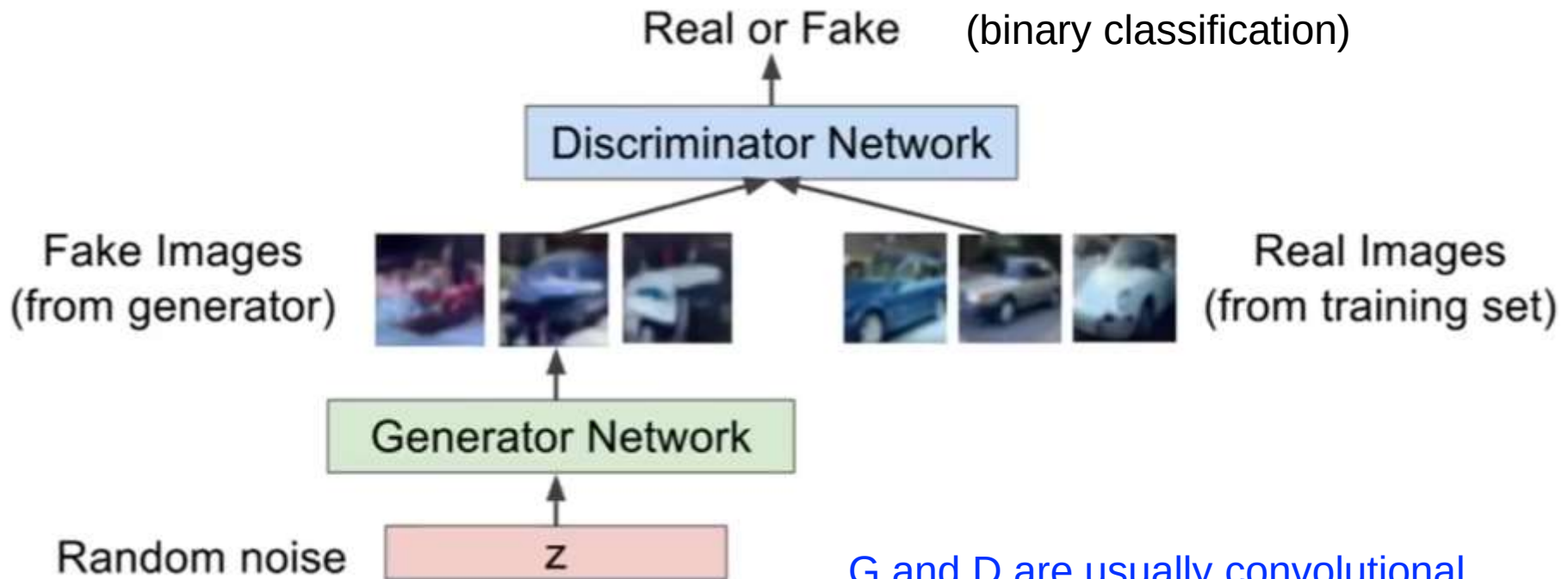
Discriminator network: try to distinguish between real and fake images (FBI agent)

Training GAN: Two-player game

Ian Goodfellow et al., “Generative Adversarial Nets”, NIPS 2014

Generator network: try to fool the discriminator by generating real-looking images

Discriminator network: try to distinguish between real and fake images



G and D are usually convolutional neural networks



Training GAN: Two-player game

Generator network: try to fool the discriminator by generating real-looking images

Discriminator network: try to distinguish between real and fake images

Train jointly in minimax game formulation:

Minimax objective function:

$$\min_{\theta_g} \max_{\theta_d} \left[\mathbb{E}_{x \sim p_{data}} \log D_{\theta_d}(x) + \mathbb{E}_{z \sim p(z)} \log(1 - D_{\theta_d}(G_{\theta_g}(z))) \right]$$

Discriminator network parameters

Generator network parameters

Training GAN: Two-player game

Minimax objective function:

$$\min_{\theta_g} \max_{\theta_d} \left[\mathbb{E}_{x \sim p_{data}} \log \underbrace{D_{\theta_d}(x)}_{\substack{\text{Discriminator output} \\ \text{for real data } x}} + \mathbb{E}_{z \sim p(z)} \log(1 - \underbrace{D_{\theta_d}(G_{\theta_g}(z))}_{\substack{\text{Discriminator output for} \\ \text{generated fake data } G(z)}}) \right]$$

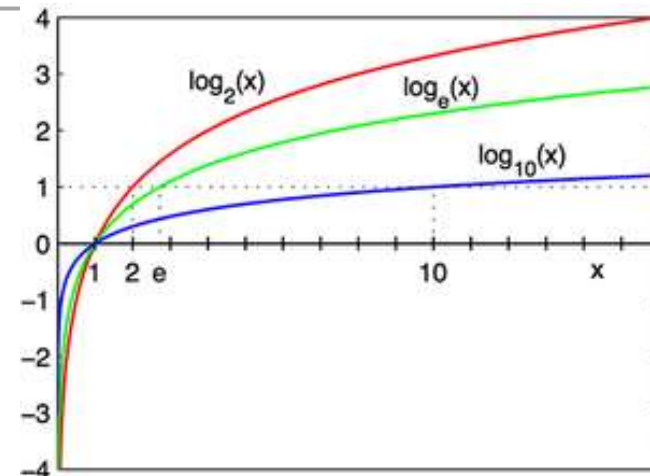
Discriminator outputs likelihood in (0,1) of real image



Training GAN: Two-player game

What is this objective trying to do?

Let's look at the discriminator first



Discriminator outputs likelihood in (0,1) of real image

Minimax objective function:

$$\min_{\theta_g} \max_{\theta_d} \left[\mathbb{E}_{x \sim p_{data}} \underbrace{\log D_{\theta_d}(x)}_{\text{Discriminator output for real data } x} + \mathbb{E}_{z \sim p(z)} \log(1 - \underbrace{D_{\theta_d}(G_{\theta_g}(z))}_{\text{Discriminator output for generated fake data } G(z)}) \right]$$



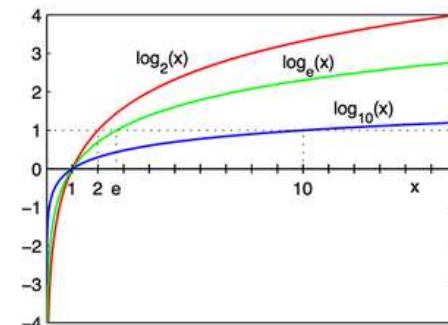
- Discriminator (θ_d) wants to **maximize objective** such that $D(x)$ is close to 1 (real) and $D(G(z))$ is close to 0 (fake)



Training GAN: Two-player game

What is this objective trying to do?

Then, let's look at the generator



Discriminator outputs likelihood in (0,1) of real image

Minimax objective function:

$$\min_{\theta_g} \max_{\theta_d} \left[\mathbb{E}_{x \sim p_{data}} \log \underbrace{D_{\theta_d}(x)}_{\substack{\text{Discriminator output} \\ \text{for real data } x}} + \mathbb{E}_{z \sim p(z)} \log(1 - \underbrace{D_{\theta_d}(G_{\theta_g}(z))}_{\substack{\text{Discriminator output for} \\ \text{generated fake data } G(z)}}) \right]$$



- Generator (θ_g) wants to **minimize objective** such that $D(G(z))$ is close to 1 (discriminator is fooled into thinking generated $G(z)$ is real)



Training GAN: Two-player game

Minimax objective function:

$$\min_{\theta_g} \max_{\theta_d} \left[\mathbb{E}_{x \sim p_{data}} \log D_{\theta_d}(x) + \mathbb{E}_{z \sim p(z)} \log(1 - D_{\theta_d}(G_{\theta_g}(z))) \right]$$

Alternate between:

1. **Gradient ascent** on discriminator

$$\max_{\theta_d} \left[\mathbb{E}_{x \sim p_{data}} \log D_{\theta_d}(x) + \mathbb{E}_{z \sim p(z)} \log(1 - D_{\theta_d}(G_{\theta_g}(z))) \right]$$

2. **Gradient descent** on generator

$$\min_{\theta_g} \mathbb{E}_{z \sim p(z)} \log(1 - D_{\theta_d}(G_{\theta_g}(z)))$$

2. **Instead: Gradient ascent** on generator, **different objective**

$$\max_{\theta_g} \mathbb{E}_{z \sim p(z)} \log(D_{\theta_d}(G_{\theta_g}(z)))$$



Training GAN: Two-player game

Putting it together: GAN training algorithm

for number of training iterations **do**

for k steps **do**

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Sample minibatch of m examples $\{x^{(1)}, \dots, x^{(m)}\}$ from data generating distribution $p_{\text{data}}(x)$.
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D_{\theta_d}(x^{(i)}) + \log(1 - D_{\theta_d}(G_{\theta_g}(z^{(i)}))) \right]$$

end for

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Update the generator by ascending its stochastic gradient (improved objective):

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log(D_{\theta_d}(G_{\theta_g}(z^{(i)})))$$

end for

Some find $k=1$
more stable,
others use $k > 1$,
no best rule.

Followup work
(e.g. Wasserstein
GAN, BEGAN)
alleviates this
problem, better
stability!

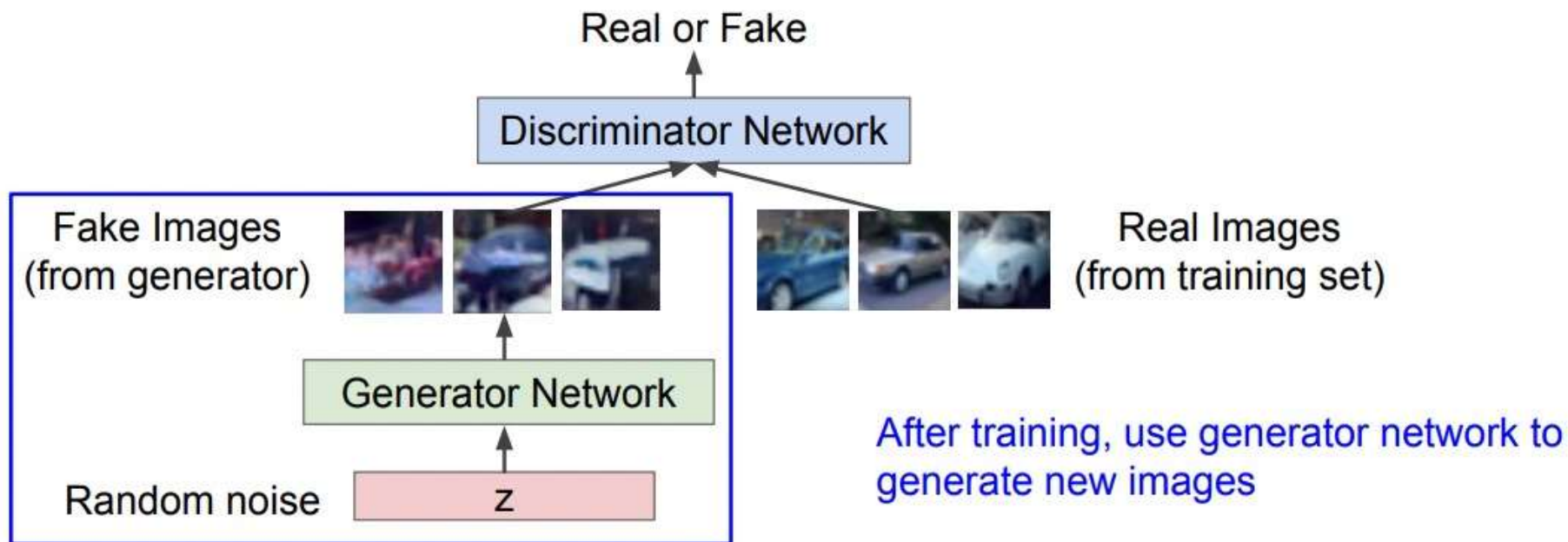
Arjovsky et al. "Wasserstein gan." arXiv preprint arXiv:1701.07875 (2017)

Berthelot, et al. "Began: Boundary equilibrium generative adversarial networks." arXiv preprint arXiv:1703.10717 (2017)

Training GAN: Two-player game

Generator network: try to fool the discriminator by generating real-looking images

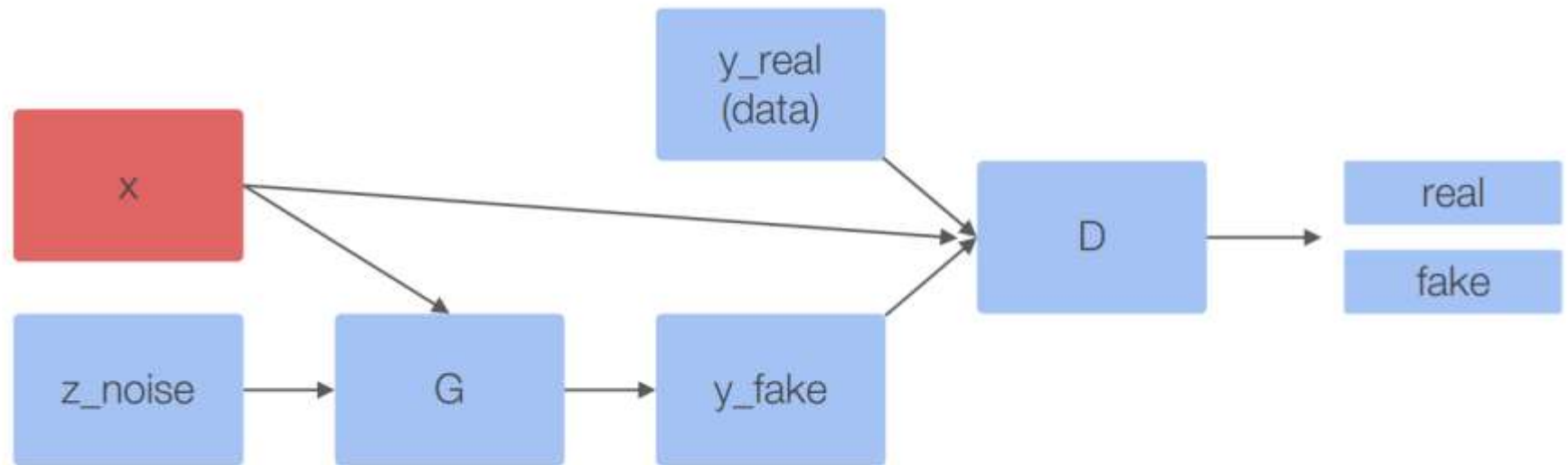
Discriminator network: try to distinguish between real and fake images



-
- <https://github.com/eriklindernoren/PyTorch-GAN>
 - How to Train a GAN? Tips and tricks to make GANs work:
<https://github.com/soumith/ganhacks>
 - DCGAN Tutorial:
https://pytorch.org/tutorials/beginner/dcgan_faces_tutorial.html
 - Video tutorial “DCGAN implementation from scratch”:
https://www.youtube.com/watch?v=IZtv9s_Wx9I&t=0s

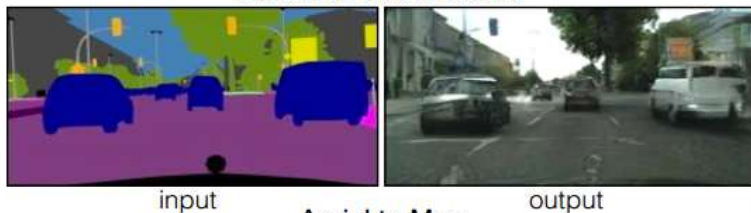
Condition GAN

In the Conditional GAN (CGAN), the generator learns to generate a fake sample with a specific condition or characteristics (such as a label associated with an image or more detailed tag) rather than a generic sample from unknown noise distribution.

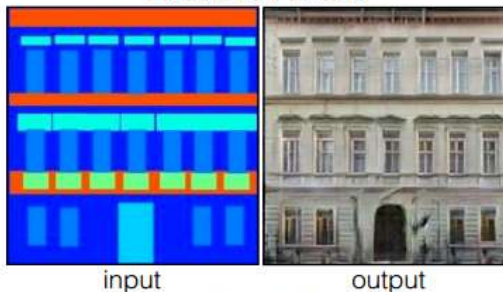


pix2pix: Image-to-Image Translation

Labels to Street Scene



Labels to Facade



BW to Color



Aerial to Map



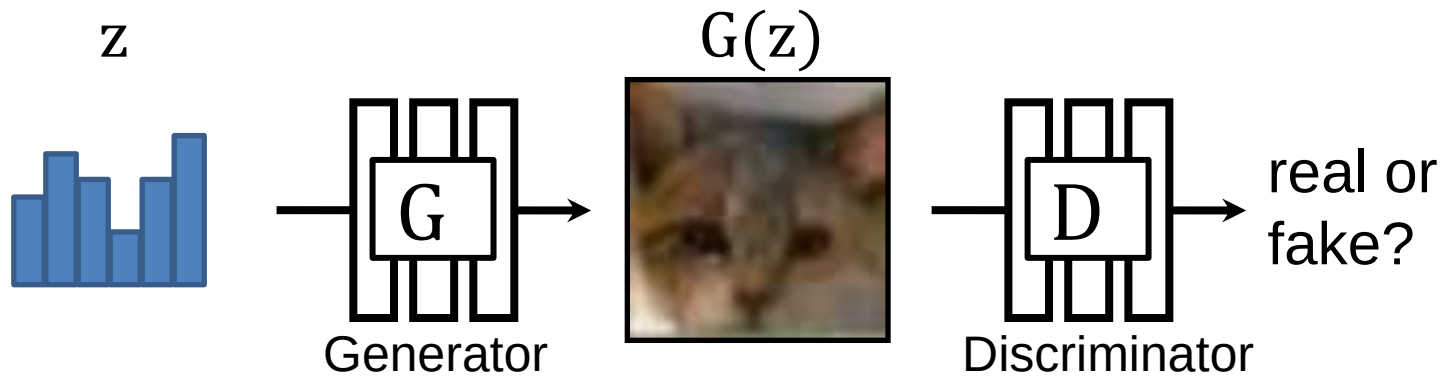
Day to Night



Edges to Photo

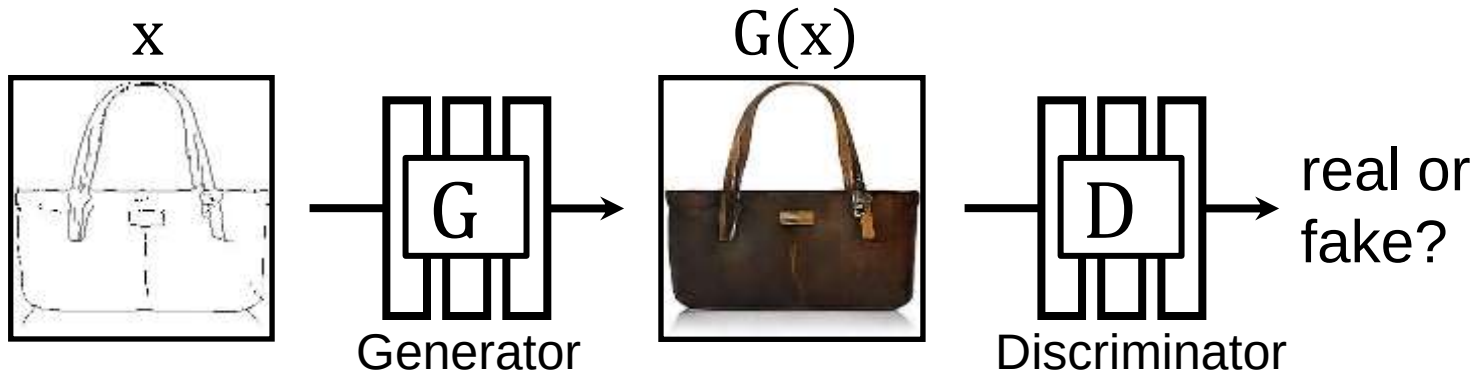


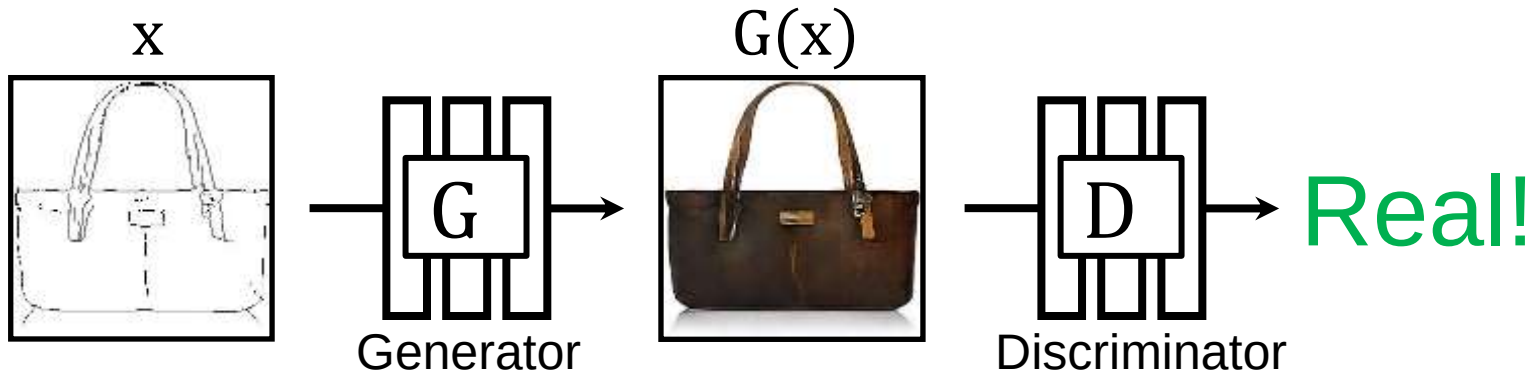
Image-to-image Translation with Conditional Adversarial Nets
Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, Alexei A. Efros. CVPR 2017



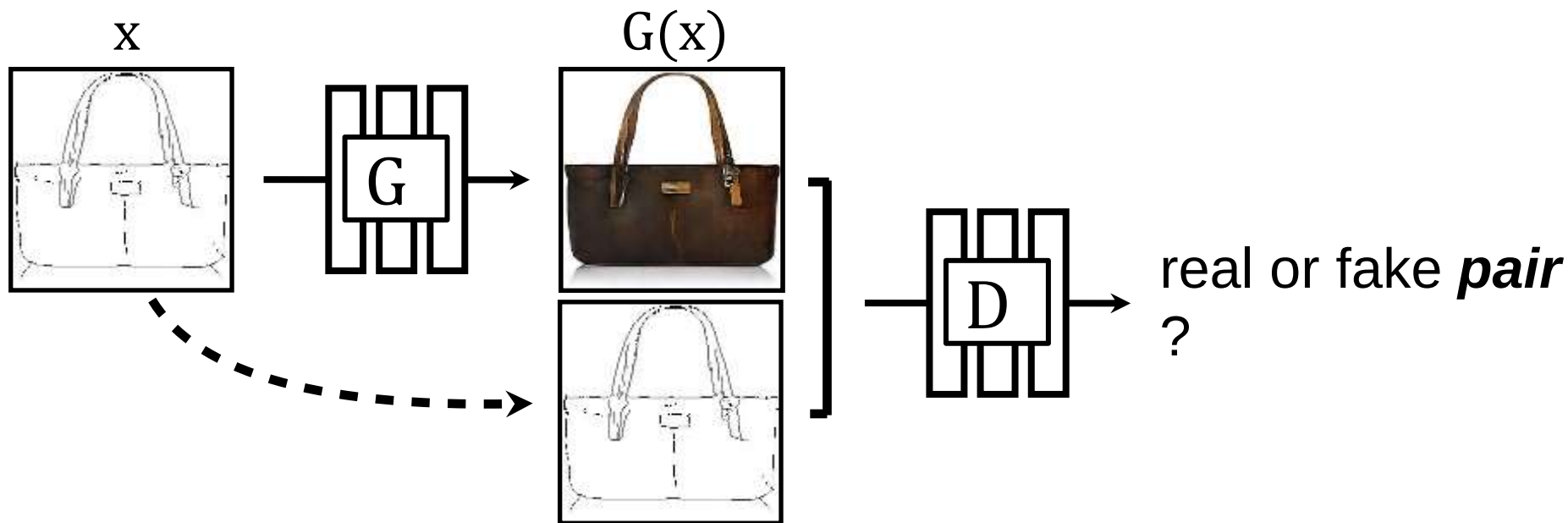
$$\min_G \max_D \mathbb{E}_{z,x} [\log D(x) + \log(1 - D(G(z)))]$$

[Goodfellow et al. 2014]





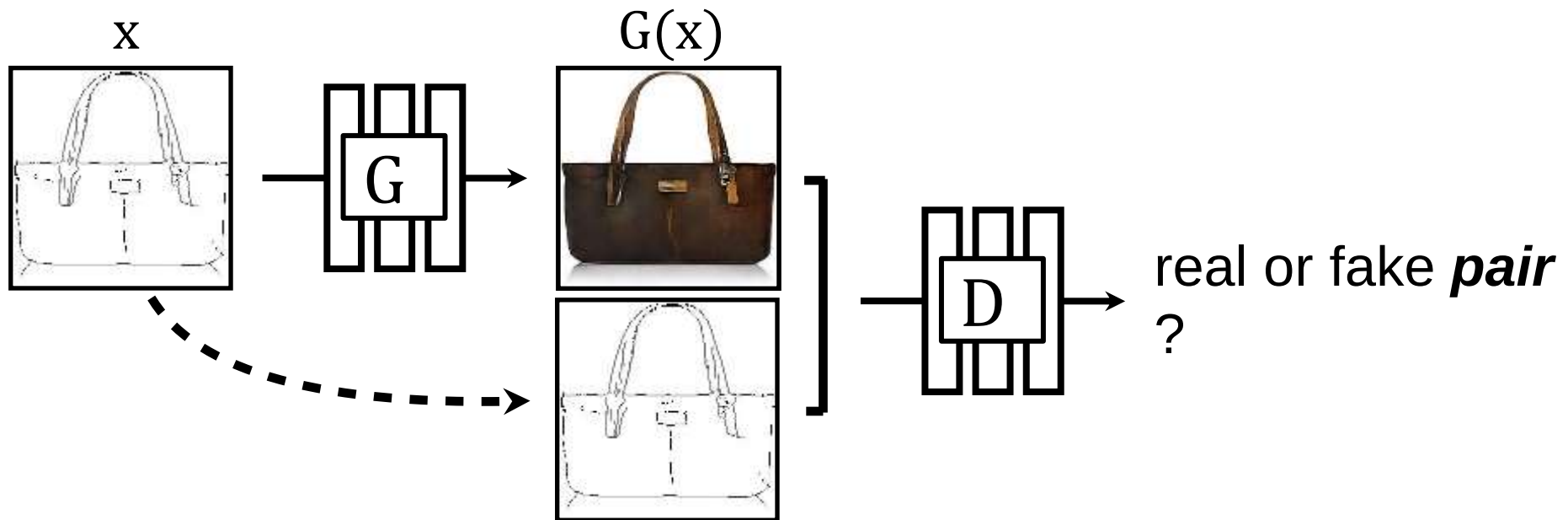




$$\min_G \max_D \mathbb{E}_{x,y} \left[\log \underbrace{D(x, y)}_{\text{real pair}} + \log(1 - \underbrace{D(x, G(x))}_{\text{fake pair}}) \right]$$

match joint distribution $p(G(x), y) \sim p(x, y)$





The generator is tasked to not only fool the discriminator but also to be near the ground truth output in an L2 sense. We also explore this option, using L1 distance rather than L2 as L1 encourages less blurring:

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y,z} [\|y - G(x, z)\|_1].$$

Our final objective is

$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G).$$

Edges → Images

Input

Output



Input

Output



Input

Output



Edges from [Xie & Tu, 2015]

Sketches → Images

Input



Output



Input



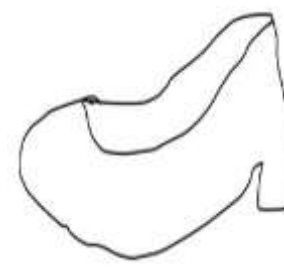
Output



Input



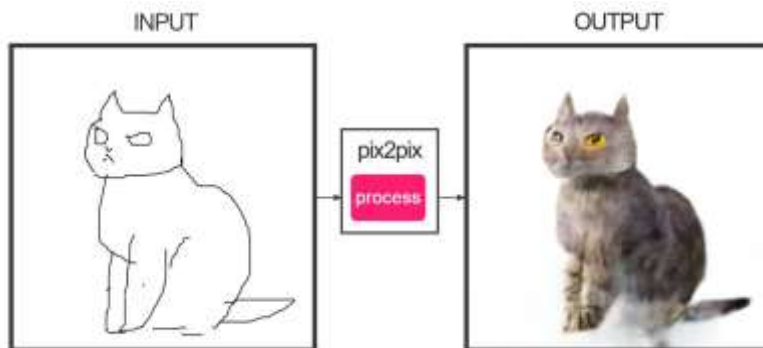
Output



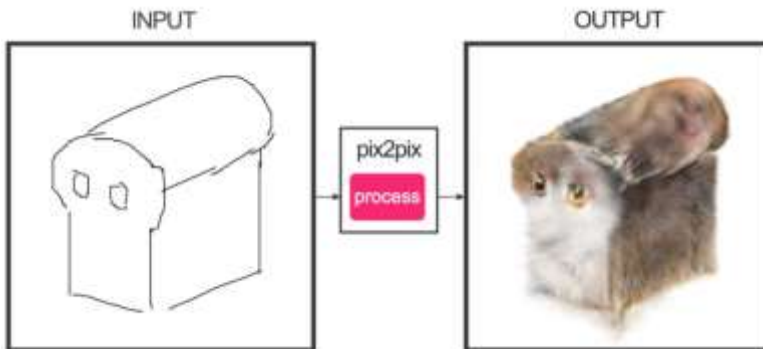
Trained on Edges → Images

Data from [Eitz, Hays, Alexa, 2012]

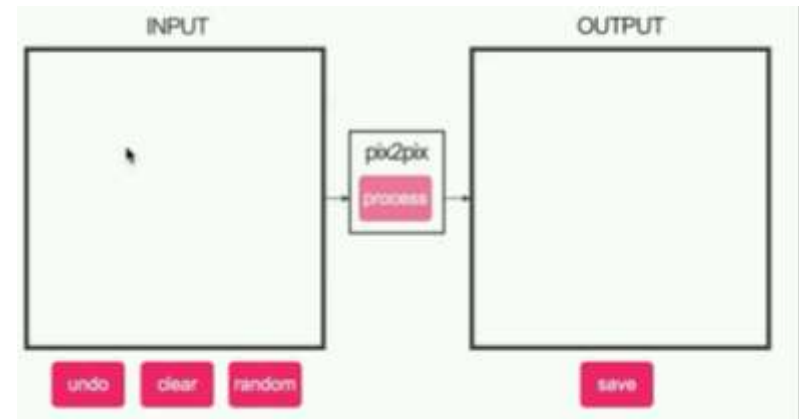
#edges2cats [Christopher Hesse]



@gods_tail



Ivy Tasi
@ivymyt



@matthematician



Vitaly Vidmirov @vvid

<https://affinelayer.com/pixsrv/>

Input



Output



Groundtruth



Data from
[\[maps.google.com\]](https://maps.google.com)



BW → Color

Input

Output

Input

Output

Input

Output



Data from [Russakovsky et al. 2015]

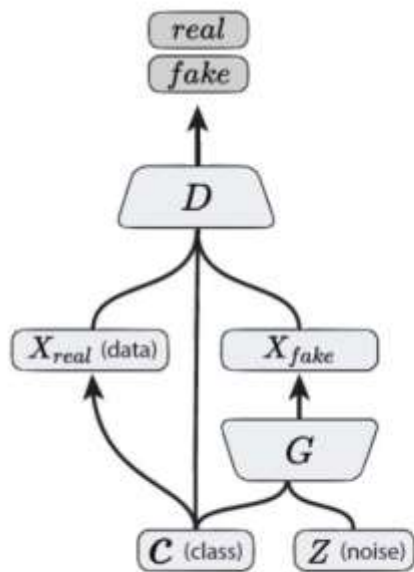
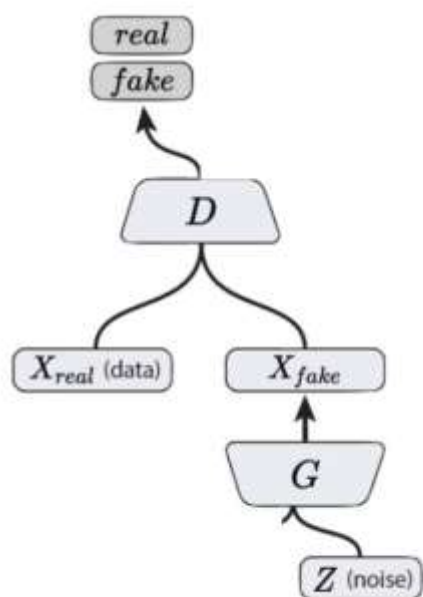
Unpaired Image-to-Image Translation using Cycle- Consistent Adversarial Networks

**a.k.a.
CycleGAN**

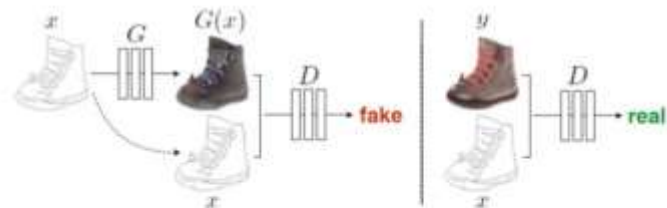
Zhu, Jun-Yan, et al. "Unpaired image-to-image translation using cycle-consistent adversarial networks." *ICCV*, 2017.



Previously, Pix2Pix



Conditional GAN



The challenge

Paired

x_i

y_i



- Expensive to collect pairs.
- Impossible in many scenarios.

Unpaired

X

Y



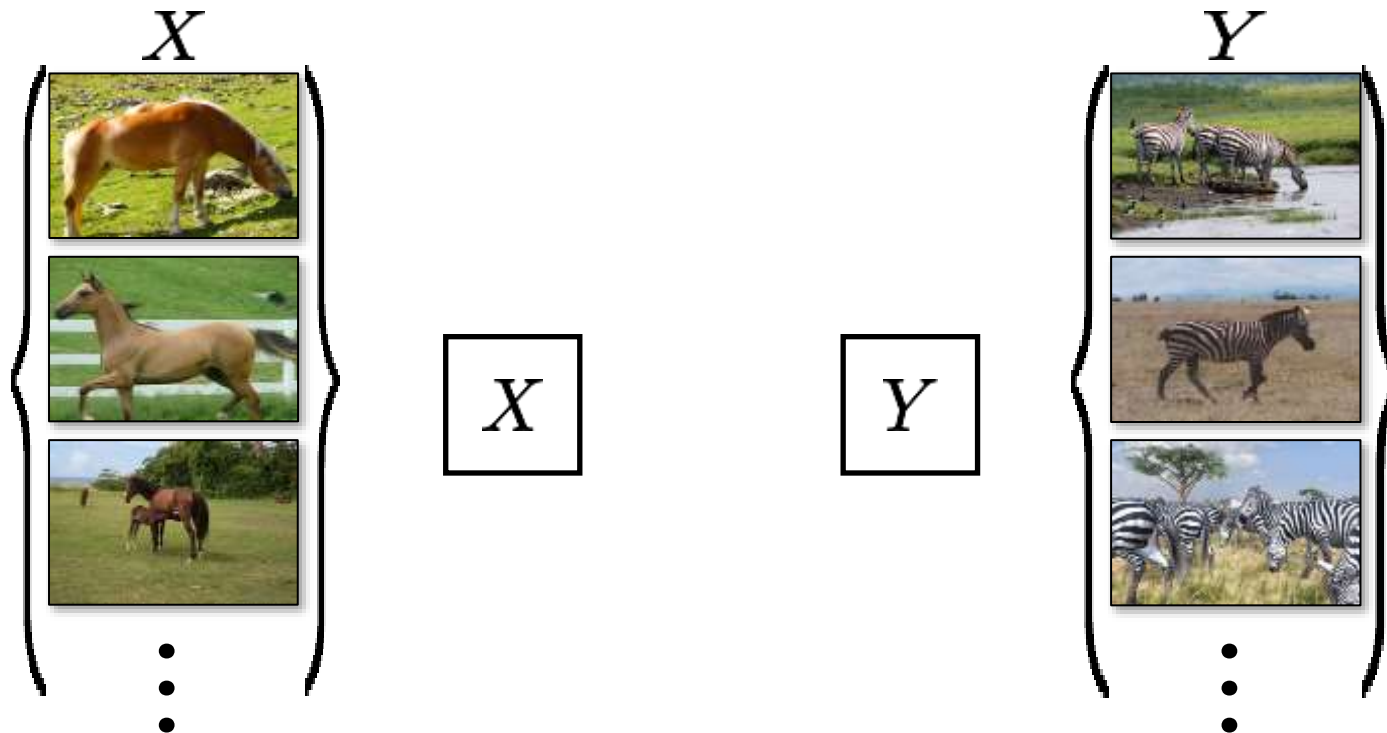
Horse $\leftrightarrow$ zebra: how to get zebras?

slides credit: Jun-Yan Zhu



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Cycle-Consistent Adversarial Networks

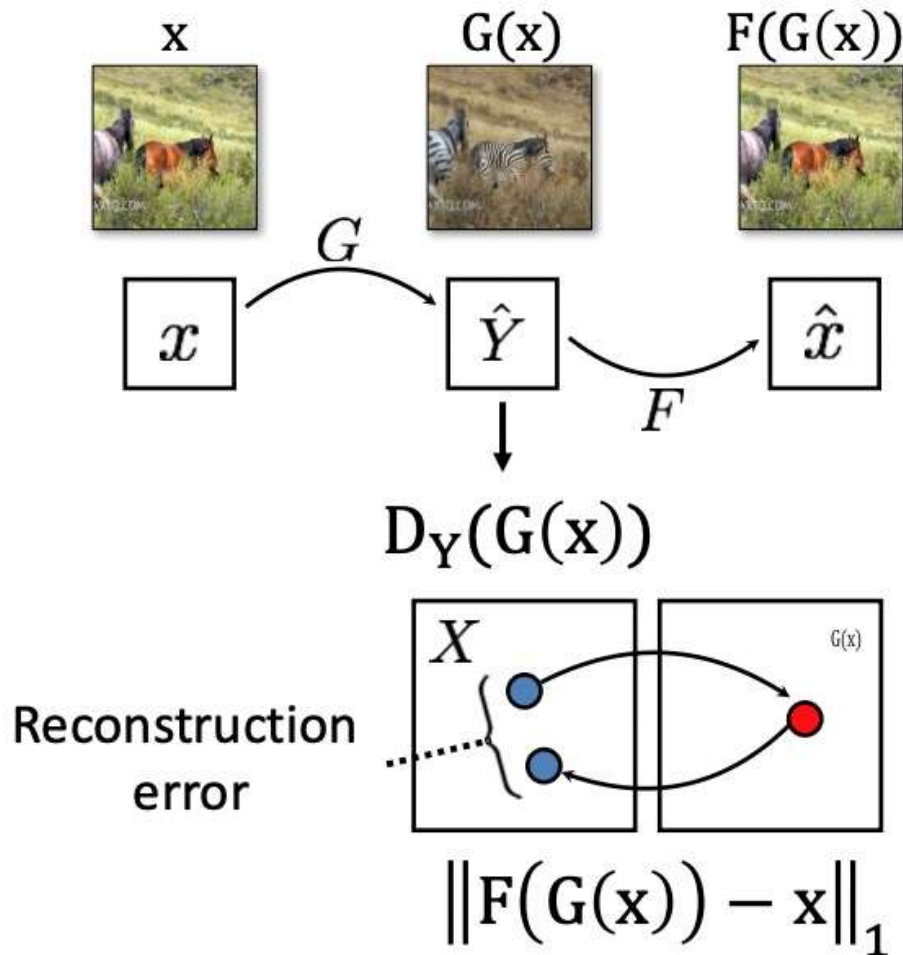


slides credit: Jun-Yan Zhu

[Zhu*, Park*, Isola, and Efros, ICCV 2017]



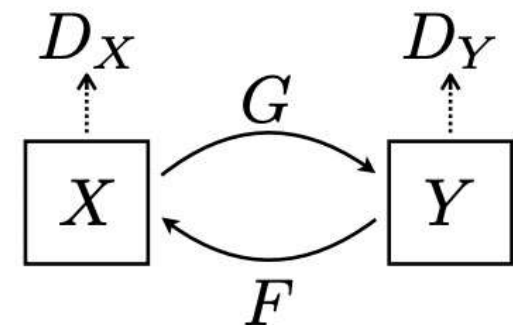
Cycle-Consistent Adversarial Networks



$$G : X \rightarrow Y$$

$$F : Y \rightarrow X$$

D_Y encourages G to translate X into outputs indistinguishable from domain Y , and vice versa for D_X and F .

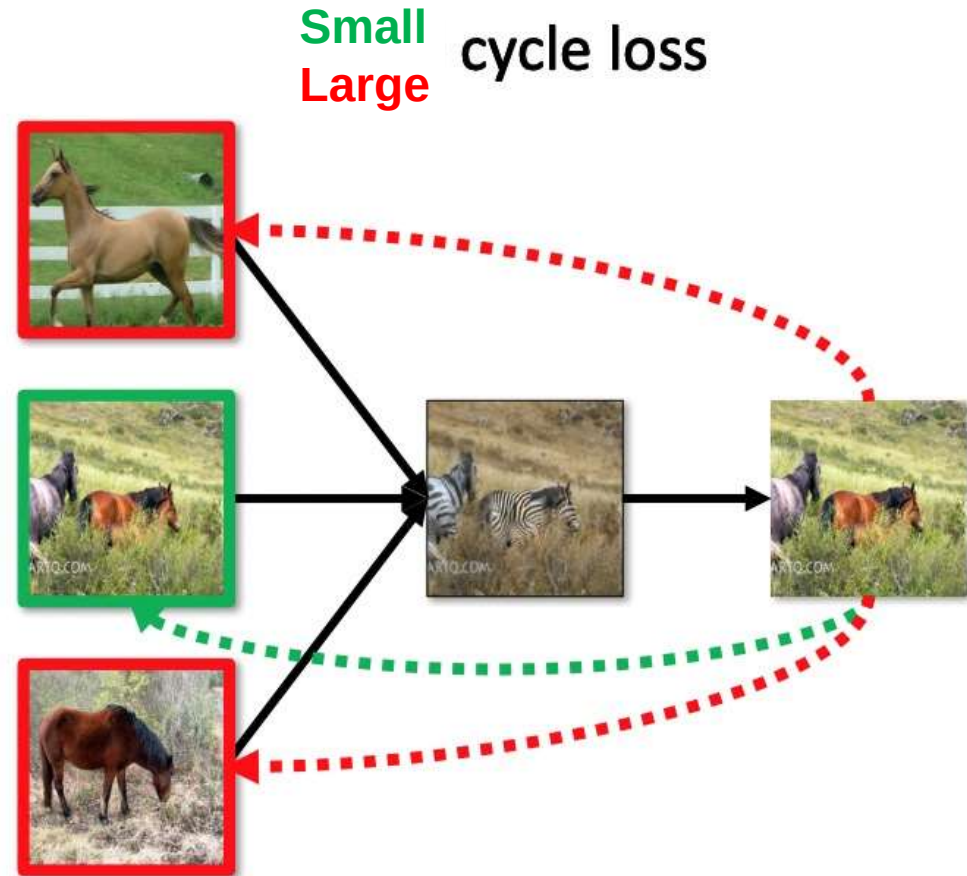
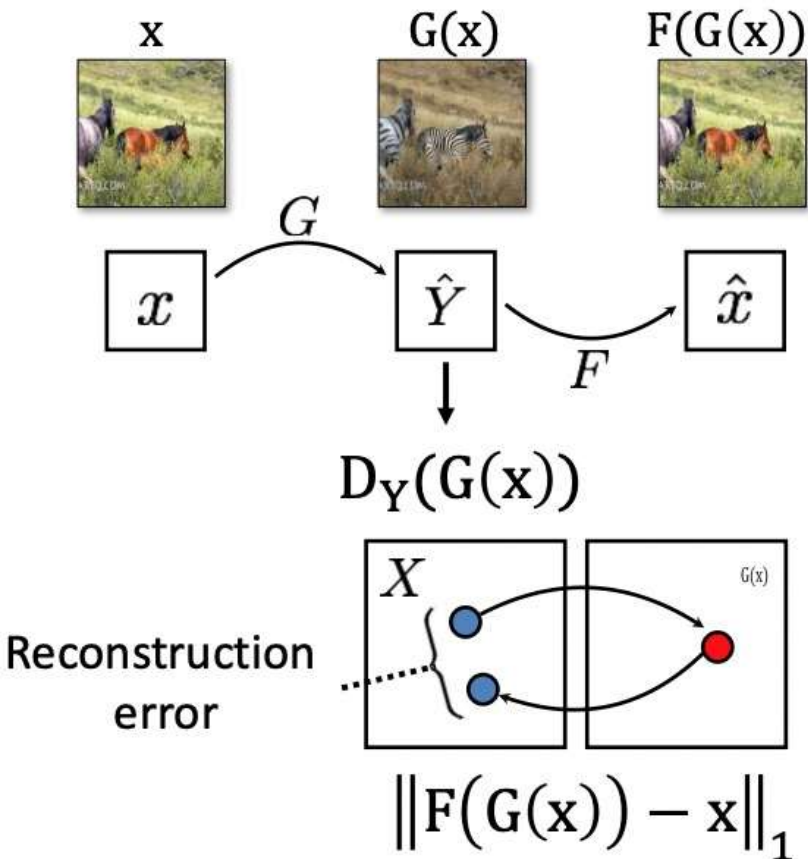


slides credit: Jun-Yan Zhu

[Zhu*, Park*, Isola, and Efros, ICCV 2017]



Cycle-Consistent Adversarial Networks

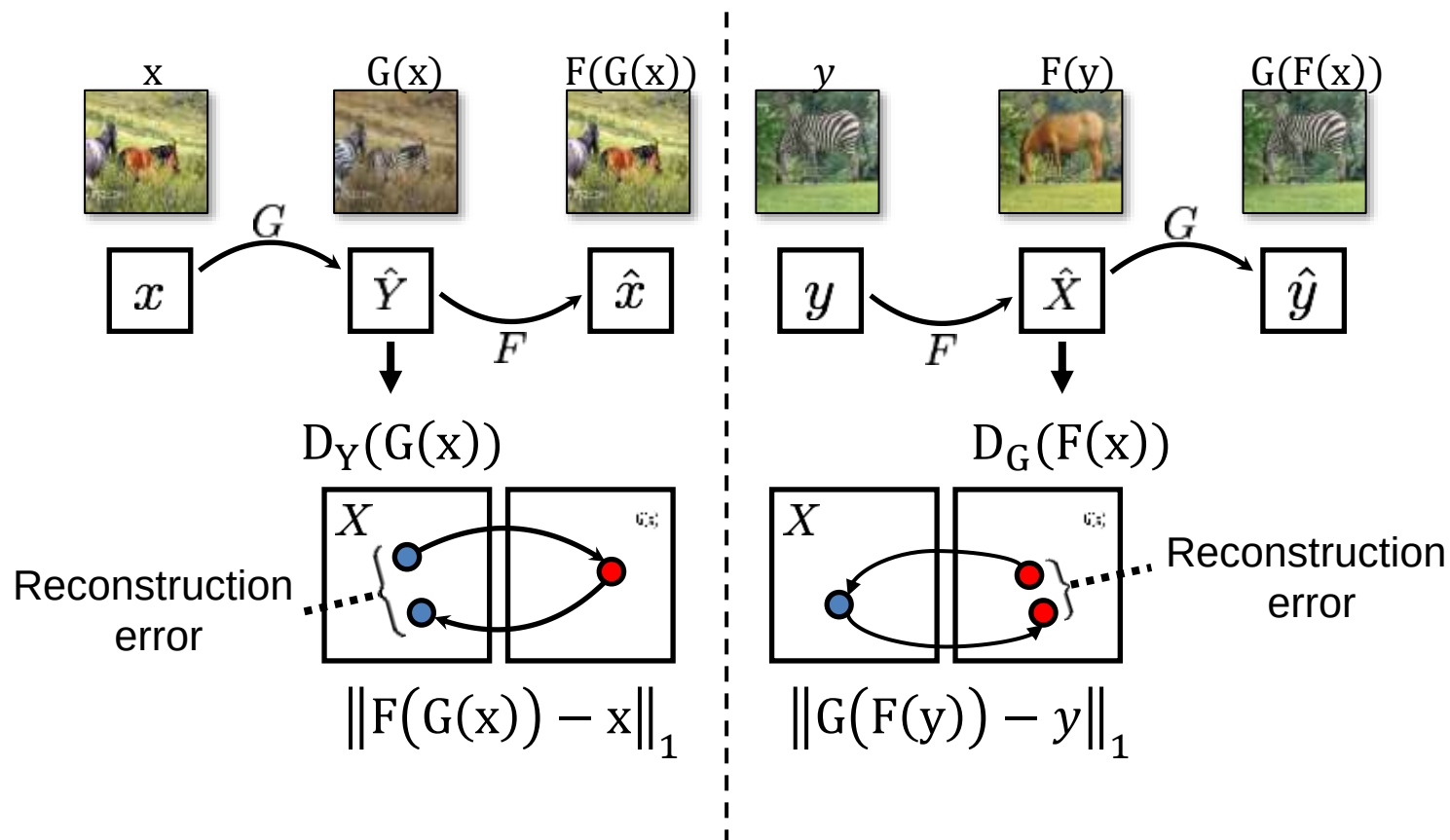


slides credit: Jun-Yan Zhu

[Zhu*, Park*, Isola, and Efros, ICCV 2017]



Cycle Consistency Loss



slides credit: Jun-Yan Zhu

[Zhu*, Park*, Isola, and Efros, ICCV 2017]



Monet's paintings → photos



slides credit: Jun-Yan Zhu

Monet's paintings → photos



slides credit: Jun-Yan Zhu





slides credit: Jun-Yan Zhu



slides credit: Jun-Yan Zhu



Failure case



slides credit: Jun-Yan Zhu



Failure case



slides credit: Jun-Yan Zhu



CycleGAN Demo



Real zebra image

Medical Imaging Applications

MR → CT [Wolterink et al] arxiv: 1708.01155



Input MR

Generated CT

Ground truth CT

- **MRI reconstruction** [Quan et al.]
arxiv:1709.00753
- **Cardiac MR images from CT** [Chartsias et al. 2017]



Example 1: Low dose CT denoising



Input

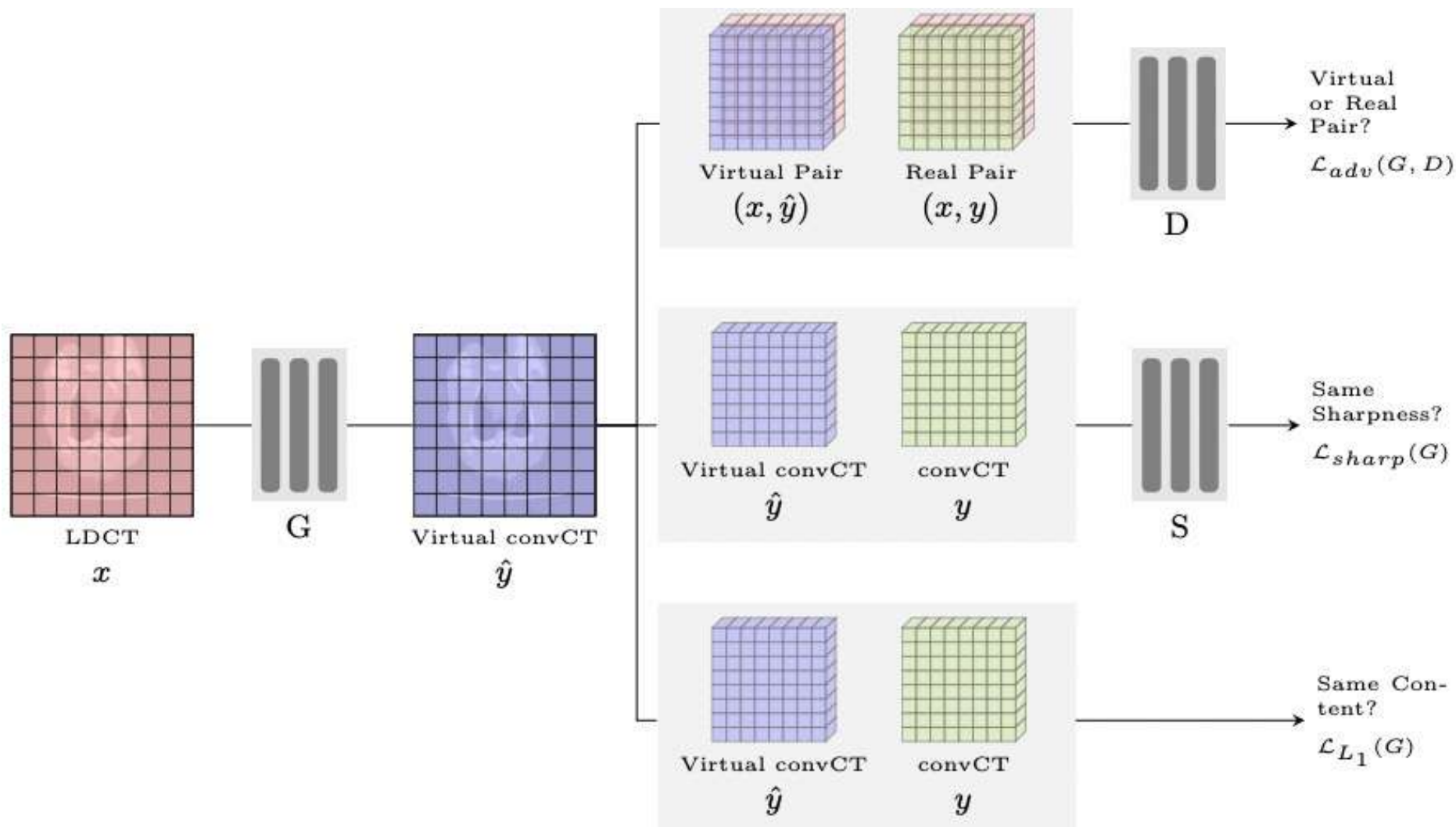


Denoised

Yi, Xin, and Paul Babyn. "Sharpness-aware low-dose CT denoising using conditional generative adversarial network." Journal of digital imaging 31.5 (2018): 655-669.



Example 1: Low dose CT denoising



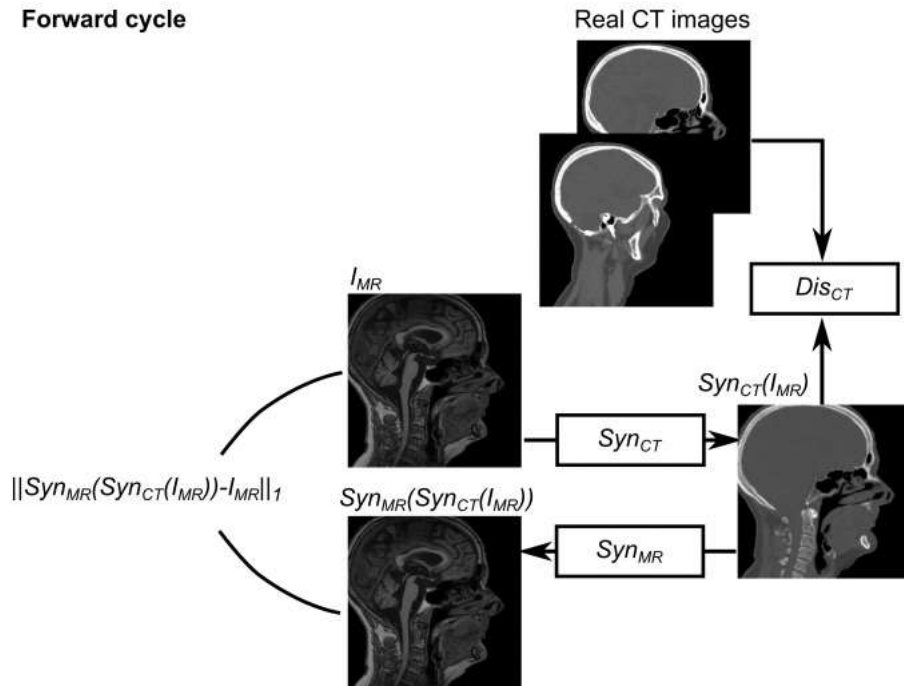
Yi, Xin, and Paul Babyn. "Sharpness-aware low-dose CT denoising using conditional generative adversarial network." Journal of digital imaging 31.5 (2018): 655-669.



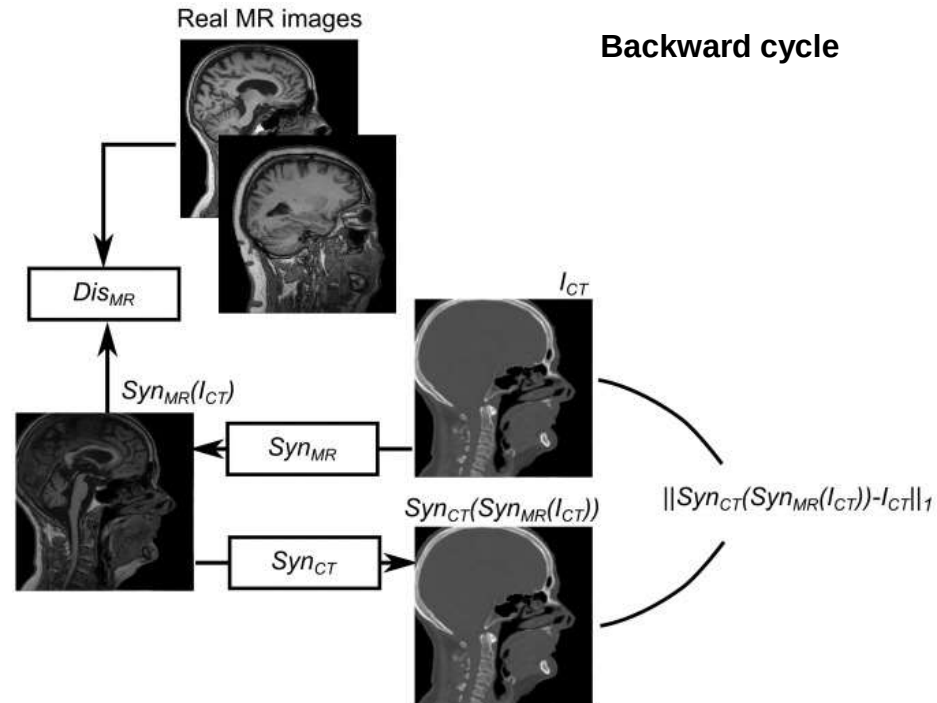
Example 2: MR to CT Synthesis

MR to CT Synthesis using CycleGAN

Forward cycle



Backward cycle



Wolterink, Jelmer M., et al. "Deep MR to CT synthesis using unpaired data." *International workshop on simulation and synthesis in medical imaging*. Springer, Cham, 2017.



Example 2: MR to CT Synthesis

MR to CT Synthesis using CycleGAN

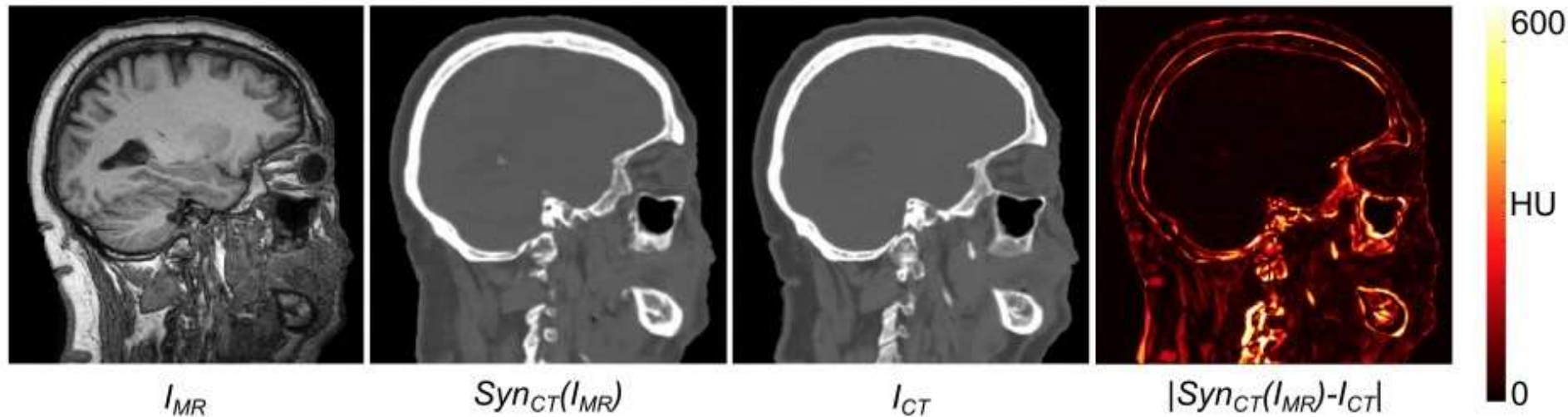
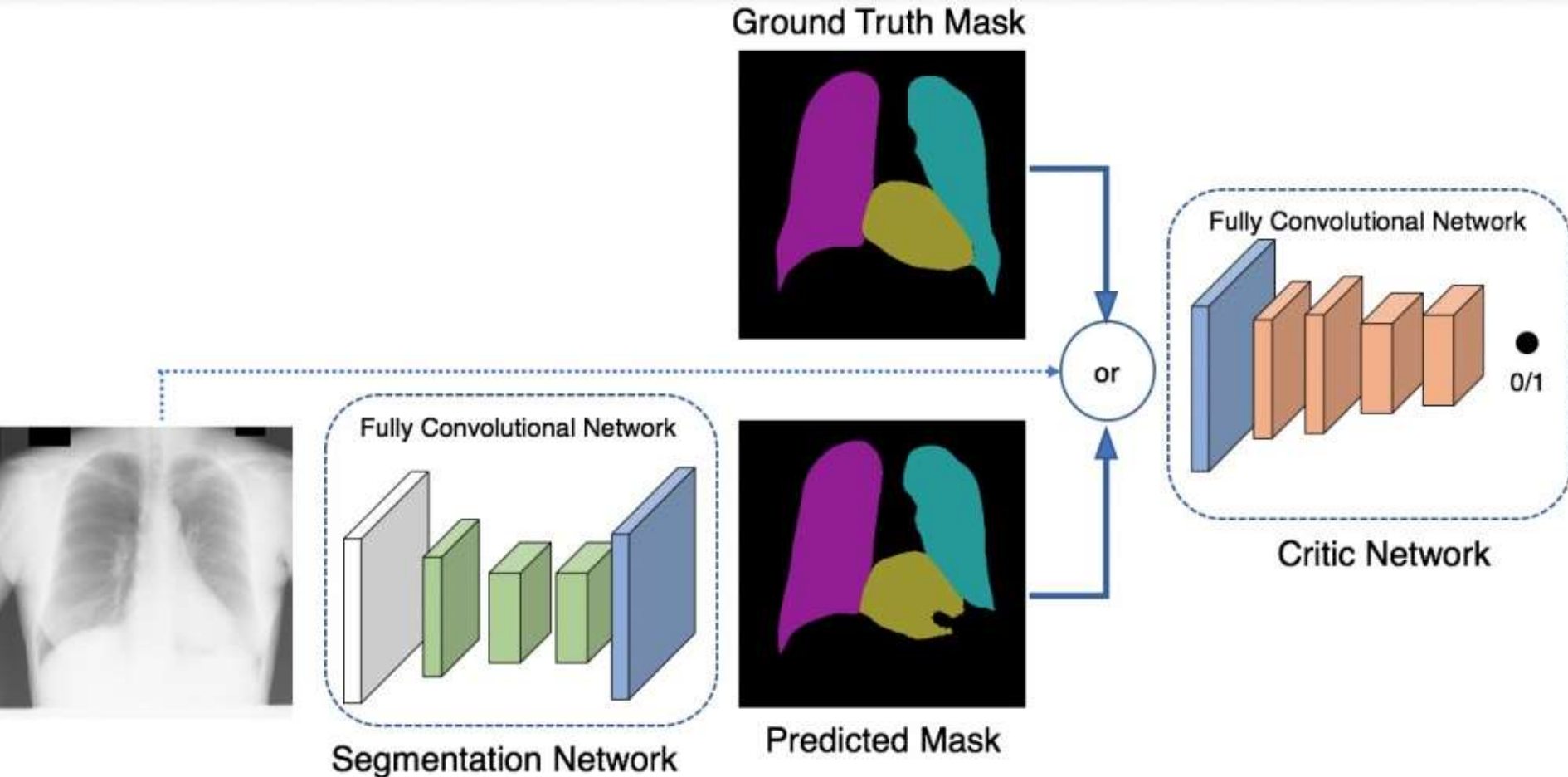


Fig. 4: *From left to right* Input MR image, synthesized CT image, reference real CT image, and absolute error between real and synthesized CT image.

Wolterink, Jelmer M., et al. "Deep MR to CT synthesis using unpaired data." *International workshop on simulation and synthesis in medical imaging*. Springer, Cham, 2017.

Example 3: Organ Segmentation in Chest X-rays

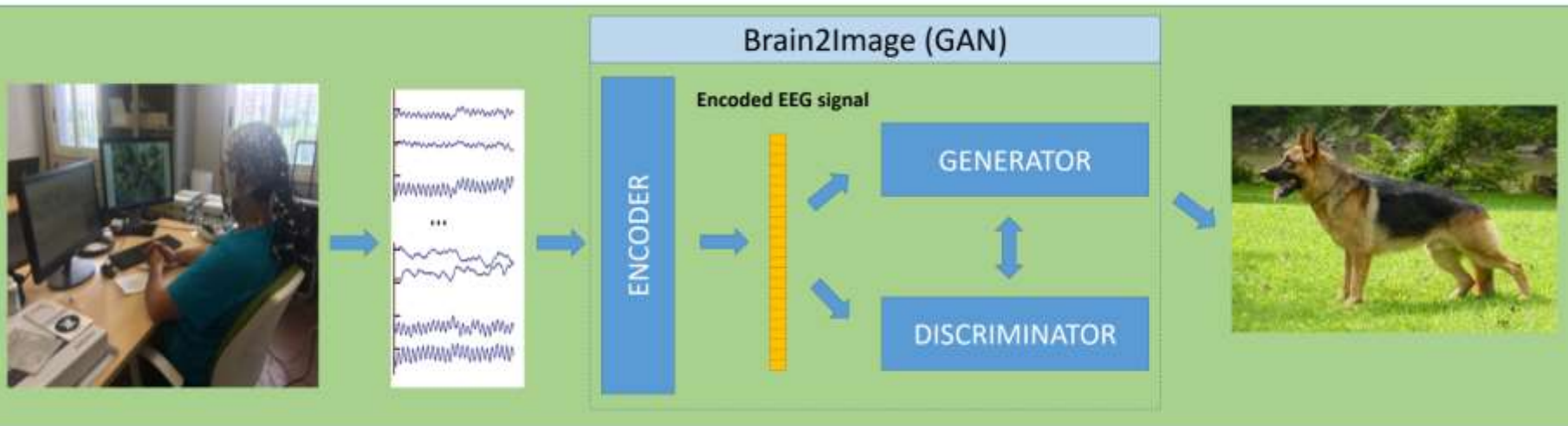


Dai, Wei, et al. "Scan: Structure correcting adversarial network for organ segmentation in chest x-rays." *Deep learning in medical image analysis and multimodal learning for clinical decision support*. Springer, Cham, 2018. 263-273.



Example 4: Brain Signals into Images

Brain computer interface

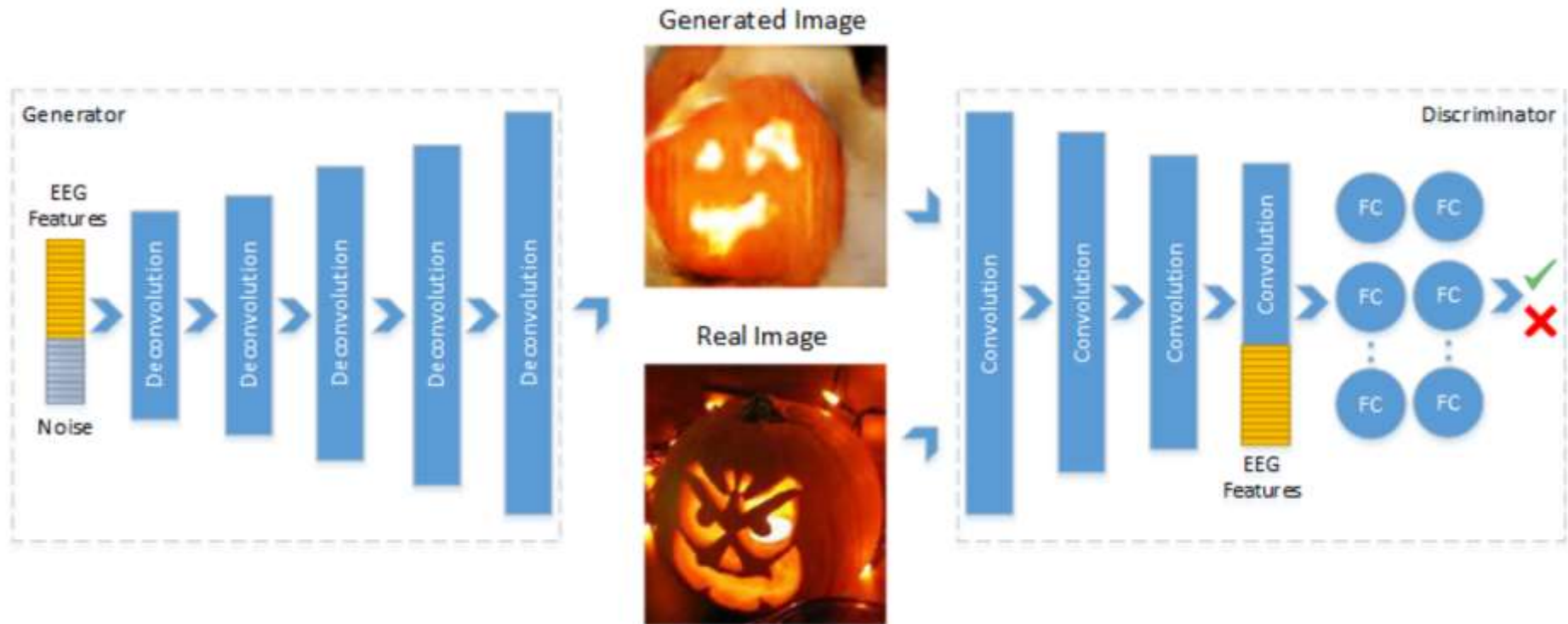


EEG-based image generation – “reading the mind”

I kavasidis, S. Palazzo, C. Spampinato, D. Giordano, M. Shah, Brain2Image: Converting Brain Signals into Images, ACM Multimedia, 2017.



Brain Signals into Images



I kavasidis, S. Palazzo, C. Spampinato, D. Giordano, M. Shah, Brain2Image: Converting Brain Signals into Images, ACM Multimedia, 2017.





(a) Airliner



(c) Panda

I kavasidis, S. Palazzo, C. Spampinato, D. Giordano, M. Shah, Brain2Image: Converting Brain Signals into Images, ACM Multimedia, 2017.



"DeepFake"

- A person in an existing image or video is replaced with someone else's likeness, usually by GAN (sometimes autoencoders)



<https://github.com/deepfakes/faceswap>

Credit: Zhangyang Wang

References

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- Wang, Zhengwei, Qi She, and Tomas E. Ward. "Generative adversarial networks in computer vision: A survey and taxonomy." *ACM Computing Surveys (CSUR)* 54.2 (2021): 1-38.
- <https://github.com/xinario/awesome-gan-for-medical-imaging>
- <https://github.com/eriklindernoren/PyTorch-GAN>
- <https://github.com/nashory/gans-awesome-applications>
- How to Train a GAN? Tips and tricks to make GANs work: <https://github.com/soumith/ganhacks>
- http://cs231n.stanford.edu/slides/2021/lecture_12.pdf
- GAN collections: <https://github.com/wiseodd/generative-models>
- <https://github.com/hindupuravinash/the-gan-zoo>



Thank you!

Question?